QUOC HUY DINH (RYAN)

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PROFESSIONAL SUMMARY

Final-year Artificial Intelligence student at the University of Technology Sydney (WAM 88.88, GPA 6.75 / 7.0, FEIT Dean's List 2026) with a prior Computer Science degree and hands-on experience across computer vision, machine learning, and full-stack software development.

Currently a Data Engineer processing drone and multispectral imagery, incoming CSIRO computer vision research intern, first author of RallyVLM (hybrid CV and vision-language model analysis of badminton), and ML lead on a licence-plate recognition project.

Comfortable taking work end to end, from data pipelines and model training through to deployed products on Vercel, Docker, and Hugging Face Spaces.

Open to internship, casual, part-time, contract, and graduate roles across AI, software, and data. Fast learner who ships strong outcomes both solo and in small teams.

EDUCATION

Bachelor of Artificial Intelligence University of Technology Sydney | Sydney, Australia | Jul 2025 to Jul 2027 (expected)

- WAM 88.88, GPA 6.75 / 7.0. Included on the FEIT Dean's List 2026 for outstanding academic achievement
- Focus on machine learning, deep learning, and applied AI systems. Database Fundamentals: High Distinction (100 / 100)

Bachelor of Computer Science Ho Chi Minh City University of Technology (HCMUT) | Aug 2023 to Jun 2025

- GPA 3.6 / 4.0. Data Structures and Algorithms (C++): A grade

EXPERIENCE

Computer Vision Research Intern (incoming) CSIRO Vacation Studentship, QCAT | Pullenvale, QLD | Nov 2026 to Feb 2027

- Summer research project developing image quality metrics for industrial stereo metrology systems

Data Engineer Nallawilli Bunjil (Bunjilview) | Brookvale, NSW | Aug 2026 to Present

- Process and analyse high-resolution drone imagery and multispectral datasets (including vegetation indices such as NDVI and NDRE) to support land condition monitoring and environmental insight delivery
- Build and maintain end-to-end data processing pipelines that ingest raw drone and geographic data, apply radiometric corrections, and produce cleaned, analysis-ready outputs for downstream modelling
- Apply geospatial tooling (rasterio, NumPy, GeoPandas, GDAL) across large raster datasets to extract vegetation health indicators and spatial features

Research Assistant University of Technology Sydney | Sydney, NSW | Nov 2025 to Sep 2026

- Researching Vision-Language Models and Generative Adversarial Networks in PyTorch, running multimodal understanding and generative modelling experiments
- Cleaned and analysed noisy multi-source OCR datasets, ran iterative model experiments, and turned findings into reproducible workflows
- Built normalisation, cleaning, and feature-extraction pipelines feeding downstream modelling across several concurrent research streams
- Contributed backend modules (TCP, OBS WebSocket, and REST API layers) for a game-research project, enabling real-time messaging and programmatic control

AI AND MACHINE LEARNING PROJECTS

RallyVLM: Hybrid CV plus VLM Tactical Analysis for Badminton First author, team of 3 | PyTorch, Ultralytics YOLOv11, Gemini 2.5 Flash, Qwen3-8B, LangChain, FAISS, Streamlit

- Built a two-stage propose-confirm pipeline splitting tactical analysis into a geometric side (YOLOv11 court-keypoint homography for pixel-to-metre mapping, player and shuttle tracking, velocity-reversal shot proposal) and a semantic side (a VLM labelling shot type, handedness, and out-of-position state)
- Trained and evaluated three YOLOv11 models: court keypoints mAP 0.95, player detection mAP 0.99, shuttle detection mAP 0.57; the batch Y-reversal shot detector reaches F1 0.80 against 30 hand-annotated ground-truth contacts on a BWF rally
- Wired the extracted shot events into a conversational RAG interface (FAISS with all-MiniLM embeddings, Qwen3-8B served locally through Ollama) behind a Streamlit UI, and benchmarked a proprietary VLM against an open-source baseline

Smart Parking and Intelligent Traffic Management (ALPR) ML team lead, team of 3 | PyTorch, YOLOv11-OBB, PARSEQ, ByteTrack, Django

- Leading the machine learning work on an end-to-end automatic licence-plate recognition pipeline: YOLOv11-OBB plate detection, a fine-tuned PARSEQ transformer OCR head, ByteTrack temporal voting, and Zero-DCE++ low-light enhancement
- Benchmarked robustness across CCPD2020 subsets (base, blur, tilt, rain, night, challenge) over five ablation configurations spanning the detector and OCR axes
- Built the real-time Django application for violation detection, plate reads, and parking-slot occupancy

Transformer Chatbot from Scratch PyTorch, Lightning, FastAPI, Hugging Face Spaces

- Implemented a decoder-only transformer from first principles, including a custom tokenizer, scaled dot-product attention, causal masking, and temperature sampling, without high-level model libraries
- Architecture of 4 decoder blocks, d_model 256, and 4 attention heads over a word-level vocabulary trained on a 40-intent dataset
- Served the model through a FastAPI backend on Hugging Face Spaces, powering the live chatbot widget on a personal portfolio site

Machine Learning and Deep Learning Models from Scratch Python, NumPy, PyTorch

- Implemented an RCNN and an attention mechanism from scratch, plus regression and classification algorithms with gradient descent and regularisation, and analysed the bias-variance trade-off and convergence behaviour

SOFTWARE ENGINEERING PROJECTS

Personal Portfolio Site Next.js 16, TypeScript, Tailwind CSS, Framer Motion, Resend, FastAPI

- Built a multi-page Next.js portfolio (App Router) with shared layout and dedicated routes for resume, projects, research, and contact, plus dark mode, responsive layout, and project filtering by industry
- Integrated a Resend-powered contact form with server-side validation and proxied chatbot requests through a Next.js API route to a separate FastAPI service
- Live in production at dinhquochuy.vercel.app, deployed on Vercel with CI on every push to main

AI-Powered Coffee Shop Manager Dashboard Python, LangChain, Redis, Docker

- Built an end-to-end AI briefing system delivering automated morning reports on revenue trends and operational insights, removing manual data review
- Developed a single LangChain pipeline covering natural language Q and A, revenue summarisation, inventory forecasting, and scheduled daily reports
- Added a Redis caching layer for repeated queries and containerised the system with Docker for consistent deployment

CurricuLLM, Parent-Teacher AI Platform Cambridge EduX Hackathon 2026, team of 4 | TypeScript, React, FastAPI, PostgreSQL

- Built an AI pipeline auto-generating personalised curriculum progress reports for parents from teacher input, plus a parent-facing chat interface with real-time multilingual translation
- Contributed full-stack across the React frontend and the FastAPI and PostgreSQL backend (REST API design, database schema), working in a four-person branch-and-review workflow

E-Commerce Web Application React, Express.js, Node.js, PostgreSQL, JWT

- Built a full-stack platform solo covering product listing, search and filter, cart, and checkout with order management
- Implemented JWT authentication with bcrypt password hashing, and designed a relational PostgreSQL schema normalised to 3NF with optimised CRUD queries

Game Automation and Streaming Integration Tool Python, Flask, FastAPI, TCP, OBS WebSocket

- Implemented a TCP layer for low-latency bidirectional client-server messaging across two repositories, integrated the OBS WebSocket protocol to react to live stream events, and exposed keyboard and mouse triggers via REST endpoints
- Separated the control plane from the event plane so the OBS integration and the input triggers extend independently

Frontend Projects React, JavaScript, HTML, CSS

- Built a React Tic-Tac-Toe with move history and undo (deployed on Netlify), a React music player, a React weather app, and responsive HTML/CSS/JS UI clones

Agent Market, Agent-to-Agent Negotiation Marketplace OpenAI WebMCP Challenge submission, team of 2 | Next.js 15, React 19, TypeScript, Java 21, Spring Boot 3.5, Azure SQL

- Built a multi-vendor marketplace where a buyer's and a seller's browser agents negotiate a multi-term deal (price, quantity, freebies, free shipping, coupons) through tools exposed on `document.modelContext` via WebMCP, with every order requiring explicit human confirmation
- Built the WebMCP tool layer and in-browser agent loop on the Next.js frontend, plus the Java/Spring Boot negotiation state machine and REST API on the backend, while a teammate owned the database schema and Azure SQL setup
- Covered the state machine, coupons, tokens, and catalog with 55 backend and 16 frontend tests; deployed frontend on Vercel and backend on Render

DATA AND ANALYTICS PROJECTS

Game Telemetry Data Analysis Python, Pandas, NumPy, Matplotlib, Seaborn

- Exploratory analysis of competitive-game telemetry: cleaning, feature engineering, and statistical analysis of player behaviour against match outcomes, surfacing performance and outlier insights

OCR Preprocessing Pipeline Python, OpenCV, NumPy, PaddleOCR

- Built an image preprocessing pipeline (adaptive thresholding, denoising, deskewing, contrast enhancement) and benchmarked downstream OCR accuracy across configurations to find the strongest setup for noisy inputs

Housing Price Data Analysis Python, Pandas, scikit-learn, Matplotlib

- EDA, missing-value handling, and statistical modelling to identify the strongest predictors of house price, using hypothesis testing, correlation analysis, and regression diagnostics

SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS
AI and Machine Learning: PyTorch, PyTorch Geometric, Lightning, scikit-learn, LangChain, FAISS, Ultralytics YOLO, OpenCV, Transformers, LLMs, Vision-Language Models, GANs, NLP, data cleaning, feature extraction, model evaluation
Frameworks and Libraries: Next.js, React, FastAPI, Flask, Express.js, Node.js, Pandas, NumPy, Streamlit, Matplotlib, Seaborn
Databases: PostgreSQL, Redis
Tools and Platforms: Docker, Git, Vercel, Hugging Face Spaces, Ollama, Jupyter, VS Code, AWS (learning)
Engineering Practice: REST API design, JWT authentication, TCP/IP, code review and pull requests, CI/CD, agile teamwork

CERTIFICATIONS AND AWARDS

- **Dean's List 2026** | UTS Faculty of Engineering and Information Technology, faculty level, certified by Professor Karen Whelan, Associate Dean (Teaching and Learning)
- **WIL to Work Certificate of Completion** | UTS Careers, 2026, signed by Eva Chan, Manager, UTS Careers
- **Full-Stack Web Development** | CoderSchool, 2024
- **Second Prize, Physics Olympiad (2022) and Scientific Research Contest (2018)** | HCMC Dept of Education, city level

AVAILABILITY

Based in Sydney, enrolled at UTS until July 2027. Part-time during semester, full-time over the summer break, on site or remote.